

COCHIN UNIVERSITY OF SCIENCE & TECHNOLOGY
SCHOOL OF ENGINEERING - DEPT. OF MECHANICAL ENGINEERING
METAL CUTTING & METROLOGY LABORATORY
(SIXTH SEMESTER B.TECH (CAD/CAM))

BEVEL PROTRACTOR

AIM : To take angular measurements, using bevel protractor, of given Vee-block, guideway of kirloskar lathe, table of slotting machining.

APPARATUS: Bevel Protractor, V Block, etc

PROCEDURE: The bevel protractor makes use of the same basic principles as the other vernier scale. On the mainscale the graduations are in 1° interval of an arc. The vernier scale occupying 46° on the main scale and is divided into 24 equal divisions. 12 on each side of a centre zero. As each side of the centre zero is marked from 0 to 60 minutes of an arc, then one of the division is equivalent to $60/12$ minutes of an arc; ie, 5 minute of an arc. Since 12 divisions on the vernier scale occupies 23° on the main scale each vernier division must be equal to $23/12 = 1.11^\circ/12$ thus the smallest measurement which can be conveniently read is $2^\circ - 1.11^\circ/12 = 1^\circ/12$ or 5 min.

To read the vernier protractor, the number of whole degrees on the main scale up to the vernier's centre zero mark is noted. Next the mark on the vernier scale which is adjacent to a mark on the main scale is found, and this is added to the number of whole degrees to obtain the reading.

A vernier protractor is a versatile instrument when used for measuring angles between faces on a component or work piece. It is capable of measuring from 0° to 360° , when used in conjunction with an accurate angle attachment. Care must be taken with the instrument particularly the blade which has a ground surface, if accurate readings are to be made.

SAMPLE CALCULATION:-

RESULT:-

INFERENCE:-

Note:-

- 1) Students have to draw neat figures of the instruments/ apparatus on the left side of each page of the fair record showing all details.
- 2) Draw the tabular columns only on left side of each page.
- 3) Keep your rough and fair records neat.

Ref. Books:

- 1) Engineering Metrology : I.C.Gupta
- 2) Engineering Metrology : R.K.Jain

AIM: TO FIND THE ANGULAR MEASUREMENT OF A GIVEN SPECIMEN USING BAVEL PROTRACTOR

ANGLE		MSR	VSR	TOTAL	ACCURATE ANGLE
	1				
	2				
	3				
	1				
	2				
	3				
	1				
	2				
	3				

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INSIDE MICROMETERS

AIM : To take inside dimensions using internal micrometer.
APPARATUS: Inside micrometers, specimen etc.
PROCEDURE: Some important parts of inside micro meters are micrometer unit, extension units, chucks or spacing collars, Handle etc Two types of inside micrometers are used. One is inside micro - meter caliper and the other one is inside micrometer with extension rods to increase the range of measurement. The body of the measuring head is made of suitable quality steel and its spindle and terminal measuring faces are made of high grade tool steel. It is desirable that the measuring face be tipped with tungsten carbide or be faced with a deposit of hard chromium. The micrometer screw has a pitch of 0.5 mm and the threads of screw and nut are truncated so as to confine contact to the flanks of the threads.

Extension rods are provided in order to enable any measurement to be made throughout the range or specification. These are made of suitable material. Chucks are intended for smaller adjustments in the range of measurement. The ends of spacing collars are well finished by lapping.

SAMPLE CALCULATION:-

RESULT:-

INFERENCE:-

DEPTH GAUGE

AIM: To determine the depth of a specimen using vernier depth gauge.
APPARATUS: Vernier depth gauge, test specimen etc.
PROCEDURE: The figure shows a vernier depth gauge. The vernier scale is fixed to the main body of the depth gauge and is read in the same way as a vernier calipers. Running through the depth gauge body is the main scale, the end of which provides the datum surface from which measurement are taken.

The depth gauge is carefully made so that the beam is perpendicular to the base in both directions. The end of the beam is square and flat, like the end of a steel rule, and the base is flat and true, free from curves or waviness.

While using the instrument, first of all, make sure the reference surface, on which the depth gauge is rested, is satisfactorily true flat and square. Measuring depth is a little like measuring an inside diameter. Measuring depth is a little like measuring an inside diameter. The gauge itself is true and square but it can be imperceptibly tipped or canted, because of the reference surface perhaps and offer an erroneous reading.

SAMPLE CALCULATION:-

RESULT:-

INFERENCE:-

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METROLOGY AND MEASUREMENT LAB

OBSERVATION TABLE

Least count= mm

Instrumental error= mm

Sl.no	MSR mm	VSR	TOTAL READING (x_i) mm	MEAN $\bar{x}$	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	$\Sigma(x_i - \bar{x})^2$
h_1							

SAMPLE CALCULATION

M.S.R =

V.S R =

Total Reading= MSR+ (VSR*LC) $\pm$ Instrumental error

= mm

$$\sigma = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{(n-1)}}$$

Random Uncertainty (RU) = $\sigma / \sqrt{n}$ =

Instrumental Uncertainty (IU) = LC/2 =

Total Uncertainty (TU) = $\sqrt{IU^2 + RU^2}$ =

Dimension = $\bar{x} \pm TU$

= mm

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SLIP GAUGES

AIM : To take some dimensions using slip gauges

APPARATUS: Slip gauges, Slip gauge accessories, etc.

PROCEDURE: Slip gauges are made from high grade cast steel to exceptionally close tolerances. There are five grades of slip gauges available as follows.

- Grade 2:- This is the work shop grade
- Grade 1:- Used for more precise work such that carried out in a good class tool room.
- Grade 0:- This is more commonly known as inspection gauges and its use is confined to toolroom or machine shop inspection.
- Grade 00:- This grade would be kept in the standard room and would be used for work of the highest precision only.
- Celebration grade:- This is a special grade with the actual sizes of the slips stated or calibrated on a special chart supplied with the set.

Wringing of slip gauge is of fundamental importance in precision measurement. A combination of sliding and twisting motions has been found to affect adherence greatly and a similar combined movement is also used to separate them. The standard method of wringing the gauges together is first to bring the faces of the gauges in to contact at right angles to one another and then turn them through 90 degrees. This method causes less rubbing of the surfaces and for this reason the wringing should not be done by sliding two gauges parallel to each other.

Slip gauge accessories:- These accessories ie, holders, jaw scribers, centre points extend the practical field for slip gauges. The use of holder becomes essential when it is desired to use the slip gauge combination of considerable length in order to ensure more rigidity. The holder comprises of two slotted side plates secured to end block whereby they may be lightly clamped in position.

The measuring jaws are always supplied in pairs. The jaws of two types - Type A and Type B. Type A jaws are used for internal measurement and type B jaws are used for external measurement. In Type B measuring jaws the important role is played only by the wringing face.

Scribers and centre points in conjunction with the holder and slip gauges constitute very accurate device for marking out purpose. The scribing and centre points are designed in a manner such that when they are used in combination the diameter of a finely scribed circle does not differ from twice the size of the slip gauge stack inserted between the points by more than $\pm 0.025\text{mm}$.

Base is made of robust construction and designed such that it remains stable when used with the longest size holder.

SAMPLE CALCULATION:-

RESULT:-

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AIM: CALIBRATION OF MICROMETER USING SLIP GAUGE

OBSERVATION COLUMN

SLNO	SLIP GAUGE MEASUREMENT		MICROMETER MEASUREMENT			ERROR	% ERROR
	COMBINATION	VALUE	BARREL READING	THIMBLE READING	TOTAL READING		
1							
2							
3							
4							
5							
6							

AIM: TO FIND THE PITCH DIAMETER, PITCH ANGLE & PITCH OF A THREADED BOLT

Profilometer

OBSERVATION COLUMN

MEASUREMENT		INITIAL READING			FINAL READING			DIFFERENCE	MEAN
		BARREL READING	THIMBLE READING	TOTAL READING	BARREL READING	THIMBLE READING	TOTAL READING		
MAJOR DIAMETER	1								
	2								
	3								
MINOR DIAMETER	1								
	2								
	3								
PITCH	1								
	2								
	3								

Initial Reading

MEASUREMENT		MSR	VSR	TOTAL READING	MEAN
PITCH ANGLE	1				
	2				
	3				

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MICROMETERS

AIM : To Take the dimension of a specimen using micrometers.

APPARATUS: Micrometers, Lathe specimen etc.

PROCEDURE: The basic components of a micrometer screw that is the spindle and the thimble can be used in a variety of ways. Before the micrometer is used to measure the size of a component it is necessary to zero the instrument. To do this the thimble is rotated until the two anvils are touching and the ratchet slips. At this point the reading is taken; this should read zero with the zero reading on the barrel and the zero marking on the thimble coincide with the barrel marking. If the reading is not zero the adjusting spanner can be used to allow the micrometer to be set zero. A small hole in the barrel is provided for this purpose.

The parts whose dimensions are to be measured are placed between the anvil and spindle of the micrometer. Turn the thimble by turning the ratchet until the ratchet slips. Now take the reading on the barrel and the thimble.

Now the dimension is $\text{Barrel reading} + \text{thimble reading} \times \text{Least Count}$

[

SAMPLE CALCULATION:

RESULT:

INFERENCE:

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h_1							

SAMPLE CALCULATION

M.S.R =

V.S R =

Total Reading= MSR+ (VSR*LC) $\pm$ Instrumental error

= mm

$$\sigma = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{(n-1)}}$$

Random Uncertainty (RU) = $\sigma / \sqrt{n}$ =

Instrumental Uncertainty (IU) = LC/2 =

Total Uncertainty (TU) = $\sqrt{(IU^2 + RU^2)}$ =

Dimension = $\bar{x} \pm TU$

= mm

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VERNIER CALIPERS

AIM :- To take measurements using vernier calipers
APPARATUS:- The vernier calipers, V-block specimen etc.
PROCEDURE:- Vernier calipers is a simple device using a fixed scale and a sliding scale called vernier scale to obtain measurement of accuracy greater than that of ordinary scale. It relies on the two scale reading difference called least count .

Least count = Value of one main scale division - value of one vernier division.

Grip the object whose dimensions are to be measured between the jaws and press the jaws. Note the main scale reading which coincide with the zero of the vernier scale. Note the first vernier reading which coincide with any of the main scale reading.

Then the dimension = MSR + VSR × LC

SAMPLE CALCULATION:-

RESULT:-

INFERENCE:-

VERNIER HEIGHT GAUGE

Aim:- To take the height of specimens using vernier height gauges.

Apparatus: The vernier calipers, the test specimen etc.

Procedure:- The principle is same as that of vernier calipers. The instrument consists of an integral base and beam, a slider with two vernier scales- one for height gauge measurement and one for outside caliper measurement, a screw, a nut, nut carrier for fine adjustment of movable jaw, locking screws for the slider. Graduations on one side of the bar are arranged for measuring heights from the bottom of the base to the scriber located on the top of the measuring jaw; graduations on the other side of the bar are used for outside caliper measurements. Graduations on the main scale are in 1 mm intervals, but the vernier scale are equally divided into 50 div over 49 mm length; ie 0.98 mm is the width of one division of the vernier scale.

Base is quiet robust to ensure rigidity & stability of the instrument. The under side of the base is relieved bearing a surface round the outside edge of a atleast 7 mm width. Beam has a section so chosen to ensure rigidity during use.

Measuring jaw and scriber are used to determine the height and scribe on the part. The test specimen is placed on the surface plate. The slider is adjusted so that the measuring jaw rests on the surface. The headings on the main scale and vernier scale are noted. Thus we can measure the height.

SAMPLE CALCULATION:-

RESULT:-

INFERENCE:-

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Least count= mm

Instrumental error= mm

Sl.no	MSR mm	VSR	TOTAL READING (x_i) mm	MEAN $\bar{x}$	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	$\Sigma(x_i - \bar{x})^2$
h_1							

SAMPLE CALCULATION

M.S.R =

V.S R =

Total Reading= MSR+ (VSR*LC) $\pm$ Instrumental error
= mm

$$\sigma = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{(n-1)}}$$

Random Uncertainty (RU) = $\sigma / \sqrt{n}$ =

Instrumental Uncertainty (IU) = LC/2 =

Total Uncertainty (TU) = $\sqrt{IU^2 + RU^2}$ =

Dimension = $\bar{x} \pm TU$
= mm

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SINE BAR

- AIM :**
- 1) To measure the external taper angles accurately.
 - 2) To develop skill for setting the job and instruments

APPARATUS: Sine Bar, Surface Plate, Height Gauge, Angle Plate, Square Blocks, Tapered Surface of work piece having good surface finish..

PRINCIPLE & PROCEDURE:

The Sine Bar is one of the most widely used instrument for precious measurement of angles. It consists of a rectangular section back of suitable grade steel having accurately ground pins of equal diameter one at each end and lying a line parallel to the axis of the bar. The distance between the centres of these pins is arranged to be a standard, either 5", 10" or 15" or 125 mm, 200 mm, 250mm, 500mm etc.

The sine bar is based on the principle that in a right angled triangle see length of hypotenuse is kept constant. The sine of different angles can be obtained simply by varying the length of the perpendiculars in ΔABC , AB is the hypotenuse and if it is rotated round the point. A different length of perpendiculars BC will be obtained using the following relation:

$$\sin \alpha = BC/AB = H/L$$

If AB is made 10" the value of sine can readily be computed merely by dividing the measured height BC by 10 in the inch units and by 254 if the height is measured in mm.

PROCEDURAL STEPS :

- (i) Clean the surface plate.
- (ii) Clean the sine bar.
- (iii) Clean the work piece, and ensure that there are no damages and burrs on the surfaces of work piece.
- (iv) If there are any burrs remove them by means of oil stone.
- (v) Place the work piece on surface plate with taper surface facing the surface plate.
- (vi) Place the sine bar on tapered surface of work piece with the rollers of sine bar in upward direction.
- (vii) Clean the base of height gauge properly.
- (viii) Mount the dial indicator on the height gauge.
- (ix) Set the dial indicator on the height point of one of the sine bar roller and put some pressure on dial indicator.
- (x) Note the reading of dial indicator and height gauge scale.
- (xi) Set the dial indicator on second roller of sine bar.
- (xii) Bring the same reading on dial indicator by adjusting the height gauge.
- (xii) Note the reading of height gauge at the highest point of both the rollers of sine bar.
- (xiv) Calculate the difference of two height gauge readings which will give the height (H) of one roller with respect to other.
- (xv) The centre distance between the two rollers is known for a standard sine bar.
- (xvi) Divide the height is step (xiv) by centre distance between two rollers. This will the sine of taper angle $\sin \alpha = H/L$
- (xvii) Using sine tables the value of taper angle can be calculated.

SOURCES OF ERROR

- (i) Improper cleaning of instruments of work piece.
- (ii) Damaged instruments and damaged work piece surface.
- (iii) Improper setting of instrument.
- (iv) Initial error in measuring instruments.
- (v) Uneven pressure at two points of reading may lead to error.

PRECAUTIONS

- (i) All the instruments should be cleaned properly.
- (ii) Any burrs and damage on work piece surface should be retrieved.
- (iii) Zero error in any instrument likely to be checked and if so correct it.
- (iv) Readings should be taken carefully.
- (v) In case of circular work piece sine bar should be clamped firmly with the angle plate.

SAMPLE CALCULATION

RESULT

INFERENCE

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SINE BAR

OBSERVATION COLUMN

Length of sine bar (L)-150mm

Appox angle(α) = 30°

DIAL GUAGE READING				LENGTH MOVED ON THE SPECIMEN	ANGLE β
	MSR	VSR	TOTAL READING		
1					
2					
3					

Final angle ,($\alpha \pm \beta$) =

